

The Unphrasable Law

What Happens When the Principle of Lawfulness Is Turned on Its Formulators

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Abstract. If every phenomenon falls under law, there should be a law of our own thinking: a formal system whose theorems are exactly what we can justifiably establish. This essay argues that if such a law exists, we can never hold it - never simultaneously possess its formulation, know that it is ours, and know that it is sound. The reason is a structural flexibility: whoever holds and trusts a rulebook can establish truths the rulebook cannot (its own consistency, by Gödel's second incompleteness theorem), and therefore exceeds any book he can hold. The argument is developed in a lay idiom - rulebooks, certificates, floors - with formal backing in appendices (Gödel, Löb, Tarski, Goodstein, Turing progressions, Reinhardt-Carlson epistemic arithmetic). Against Lucas and Penrose, nothing anti-mechanist follows: mechanism is possible but unknowable in the first person. The argument extends to humanity as a collective knower, which, unlike the individual, has no "outside." Finally, the only entity that could truthfully declare its own reliability is one whose saying constitutes reality - so the intuition of total lawfulness was always, in disguise, a theological one.

A. The Intuition

At the heart of modern science, and of rationalism in general, sits one intuition: every phenomenon can be explained by law. What looks chaotic - the weather, a plague, a collapsing star - turns out, when you look deep enough, to be the working of laws that can be stated precisely. This intuition has proven itself so many times that it has become almost a definition of what we call "understanding": to understand a phenomenon is to find the law behind it.

But the intuition applies, apparently, to us as well. The human brain is also a natural phenomenon; thinking, mathematical proof, insight - all of these should fall under some law, if only because they happen in a physical system. And if so, there should exist a law - a complete rulebook - that describes our thinking capacity precisely: a list of rules whose consequences are exactly what we are capable, in principle, of establishing.

The claim I want to prove in this essay is not that no such law exists. The claim is subtler, and to my mind more disturbing: **if such a law exists, we will never hold it.** And the reason is astonishingly simple: we are flexible enough to override any rulebook placed before us. And what we override is not a description of us.

That is the whole essay, in one sentence. The rest is detail: why the override is not a metaphor but a mathematical theorem; what exactly it proves and what it does not; what happens when you try to escape it into the collective - into humanity as a whole, as one knowing body, which unlike the individual has no "outside"; and finally - what could possibly stand at the top of this ladder. There, I will argue, a surprise is waiting: the answer is not a book.

B. The Simple Idea

Here is a thought experiment. Every scrap of information about me has been collected - everything I ever said and wrote, everything I did, every scan of my brain. A computer has processed it all, and now it informs me: "According to all the data - here is what you will do next. Here is what you will assume. Here is what you will invent." What will I do? Something else. Not because there is some mysterious core in me that the data misses, but because the announcement itself is a new piece of data: the moment the prediction is handed to me, I can take it and act against it. To come out true, it would have had to include my reaction to it, and then my reaction to that reaction. A prediction delivered to its own subject chases its own tail.

And the same structure repeats, one level deeper, when the subject is not actions but truths. Hand me a rulebook and tell me: "This is you. This is how you think. What follows from these rules is exactly what you will ever be able to establish." What will I do? I will open it. Study it. Examine it. And if I become convinced that it is reliable - I will stand on it and establish truths that the book itself, as we will see in the next section, can never establish. I have exceeded it. It is no longer me. And I needed no trick: the trust itself was the override.

And this is true of every book. Not because I am especially clever, but because that is how the stance itself is built: whoever holds a book and trusts it sees, by that very fact, more than the book sees - just as whoever hears a prediction about himself is no longer the man it was made about. The holder stands, relative to his book, one floor up. And from here the conclusion, which is the entire claim: **if I am flexible enough to override any book I can hold, then the book that truly describes me is, by definition, a book I will never hold.** Not because it is far away, not because it is complicated - because being held is the one thing it cannot survive. The moment it is in my hand and I trust it, I am already bigger than it; and what I am bigger than cannot be my measure.

Note the condition, because everything hangs on it. In the computer experiment it was enough that the prediction be delivered to me and speak plainly; in the truths version more is needed: I override only a book I trust. A book I do not know to be reliable - I cannot stand on it; I can only guess about it. So the claim is not that there is no law, nor

that we could never stumble upon it or guess it correctly. The claim is that we will never hold the complete package: the law in hand, the knowledge that it is ours, and the knowledge that it is reliable. One of the three will always be missing - because if all three held, I would override the law by the very act of holding it, and it would cease to be my law.

The flexibility, in other words, is not evidence that we have no law. It is evidence that our law - if there is one - is built so that it slips out of our hands at exactly the moment we think we have caught it.

C. What Gödel Proved

The override I described - standing on a book and establishing what it cannot - is not a pretty metaphor. It is a mathematical theorem. In 1931 Kurt Gödel proved his two incompleteness theorems, and to understand what they say you need no formulas. You need only the book.

Imagine a book that collects all the rules of proof of some orderly domain of thought - mathematics, say. Three modest properties are required of it: that it does not contradict itself; that one can check mechanically, without judgment, whether a given proof actually follows its rules; and that it knows basic arithmetic - adding and multiplying whole numbers. Three minimal demands. Anything we would be willing to call "orderly science" meets them.

Gödel proved that every such book misses something: there exists a true mathematical claim that the book cannot prove. That is the first theorem. And here is the point most popular discussions miss: the book itself can know this about itself. It can prove the claim "if I do not contradict myself, there is a claim I cannot prove." It sees the shape of its own blind spot. What it cannot do is verify that the condition actually holds.

Because that is the second theorem, and it is the graver of the two: the book cannot prove that it does not contradict itself. The one thing we would want from any rulebook - "you can rely on me" - is exactly the thing no book can give. Worse: a book that declares itself reliable proves, by the very declaration, that it is not reliable. The certificate of integrity a book issues to itself is evidence against it (why? the derivation, step by step, is in Appendix A). Conclusion: a valid certificate of reliability is always signed one floor above the book - never on its own pages.

And it is not only full trust. A later theorem, Löb's theorem, shows that the book cannot establish even partial, conditional trust in itself - "if I have proved something, it is true" - with respect to any question not yet settled. Self-trust is not hard to obtain; it is impossible in a provable way. And who proves the impossibility? The very faculty of thought whose reliability is in question.

And there is a still deeper prohibition, discovered by Alfred Tarski: the book cannot even phrase its own full certificate. The modest declaration - "I do not contradict myself" - the book can say, and is only forbidden to prove. But the full declaration - "everything written in me is true" - it cannot even say: no sufficiently rich language contains the concept of its own truth. With Gödel, the book is forbidden to sign; with Tarski, it turns out the signature line cannot be printed on its pages at all. The reliability of a floor is stated, always, only on the floor above it.

And here the circle closes with the previous section: what the predicting computer experiences facing the man who breaks its forecast, Gödel proved on the blackboard. Whoever holds a book and trusts it stands where the book cannot reach: he sees that the book is reliable, and sees what follows from that reliability which the book itself cannot conclude. The override is not a rare talent and not a stroke of brilliance; it is a built-in feature of the very stance in which a book lies before someone who trusts it. My flexibility over the book is not an extra power that I have and it lacks; it is the standing itself, on the floor above it - the floor where its certificate is signed. Give me a book and trust - and I will point to truths the book misses.

Here a suspicion presses itself. The sentence Gödel built is a sentence that talks about itself - logical engineering designed in advance to break the system. Perhaps the whole thing is a logicians' trick? The suspicion has two answers, and the first is the main one: whoever overrides the book needs no trick at all. The moment I trust a book, the override is already in my hand - I say "this book is reliable," and that is the whole act. Notice that in my mouth this sentence does not talk about itself at all: it is a plain claim about an object lying in front of me, like "this table is stable." It becomes self-referential, and paradoxical, only when you try to force it into the mouth of the very book it talks about. The trick is not in the sentence; it is in the position. And Gödel's engineering was not meant to create the gap but to prove, from inside mathematics itself, that the gap exists: the famous sentence is merely the book's way of saying, in its impoverished language, what the holder says with no effort at all.

And the second answer: what lives in the blind spot is no trick either. Perhaps the missing corner houses only clever sentences that no one but logicians ever meets, while real mathematics - the numbers, the sequences, the shapes - is whole and healthy? No. In 1944 Reuben Goodstein stated a simple theorem about numbers: take a number, write it in a certain way, raise the base, subtract one, and repeat the process. The theorem says that every such sequence, even if it first leaps to fantastic heights, ends at zero. There is no self-reference here, no paradox, no "I" - arithmetic, almost school arithmetic. The theorem is true, and was proved with stronger tools. And in 1982 Kirby and Paris proved that the standard rulebook of arithmetic cannot prove it. Gödel's sentence was the stress test, engineered to expose the boundary; Goodstein's theorem is what comes into view once the boundary is exposed - that beyond it lies perfectly

ordinary mathematics too. The blind spot is not a corner. Ordinary arithmetic truths live there.

It is also important to say what Gödel did not prove. He did not prove that there are things we will never know: any book can be replaced by a stronger book that proves what its predecessor missed. What Gödel proved is that you cannot hold all three things at once: a book that is mechanically checkable, that knows arithmetic, and that misses nothing. Pick two.

(The precise statements - the two theorems, the three conditions, Löb's theorem, Tarski's theorem and Goodstein's theorem, and the dictionary between them and the language of the book, the blind spot and the floors - are in Appendix A.)

D. The Full Argument

Now the argument can be stated in full. Assume three things:

The first assumption - there is a law. My thinking is a natural phenomenon, so there exists a rulebook that describes it precisely: what the book proves is exactly what I am capable of establishing.

The second assumption - it can be found. I am able, at least in principle, to hold this book in my hand: to write out its rules, and to know with certainty that the book I have written is indeed me.

The third assumption - it is reliable. I know that my book does not lead to falsehood: what it proves - is true.

Each of the assumptions, on its own, sounds plausible. But the three together place me exactly in the stance of Section B: book in hand, knowledge that it is me, trust in it. And then the flexibility does its work - I stand on the book and establish its reliability, the very thing that it, by Gödel, cannot establish. I have established something my book does not establish; the book does not exhaust me; the first assumption has collapsed. The override has swallowed the description. In the language of floors: the three assumptions require me to be the floor above myself - and there is no such floor. (The precise derivation, step by step, is in Appendix B.)

The three assumptions cannot live together. That is the theorem. But the theorem does not say which of them is false.

And here is exactly where John Lucas went wrong in his famous paper (Lucas 1961), and after him Roger Penrose in a more sophisticated version (Penrose 1989; 1994): both read the flexibility as evidence that the first assumption is the one that falls - that there is no law, that human consciousness exceeds every machine. But that inference forgets the condition. The override needs trust: I override only a book that I know to be me and

trust. Lucas and Penrose quietly assumed that if we were a machine, we could identify which machine we are and rely on it - and only then does the override arrive and contradict. And why assume that? A machine runs its code with no guaranteed access to the source printout. This critique was hurled at Lucas already in the sixties (Putnam 1960; Benacerraf 1967) and sharpened against Penrose (Chalmers 1995; Feferman 1995): the anti-mechanist argument assumes its own conclusion.

The cautious conclusion runs in the opposite direction: we do not know whether there is a law. What we do know - with mathematical knowledge - is that if there is a law, we will never hold the complete package: we cannot both formulate it and know its validity. Lawfulness, applied to itself, does not refute itself - it **vanishes** itself: the very flexibility it produces is what guarantees that its own instance, for us, will never be presented.

This thesis has a precise vindication, obtained in frameworks built specifically to test it: the claim "I am a machine" is consistent with everything an ideal knower could ever know; the claim "I know which machine I am, and it is reliable" is refutable. As a slogan: **mechanism is possible - but unknowable in the first person.** (Details in Appendix C.)

And Gödel himself? He was more careful than Lucas. In a late lecture (Gödel 1951) he set up two possibilities of which one must be true - what became known as "Gödel's disjunction": either the human mind infinitely surpasses every finite machine, or there exist questions about whole numbers that no future course of human mathematics will ever settle. He insisted that the theorems prove the either-or, but neither of its horns. Our argument here is, if you like, a sharpening of the second horn: if we are a system, the first-ranking unsettleable question is our own consistency - and the precise contour of our limits is a fact we can believe in, but cannot hold.

E. Two Necessary Refinements

The first refinement: the barrier is first-person. Notice what the argument does *not* forbid. It does not forbid someone else from holding my book. A brain researcher, a system stronger than me, an outside observer - all of these can, in principle, write out my rulebook in full, verify that it is reliable, and see from outside the claim I will never establish. And when they do, they too override my book - but no one claimed my book describes them. The override contradicts the description only when the overrider and the described are one. The very same sentence - "this book is reliable" - is valid in the holder's mouth and invalid in the book's: the certificate is disqualified not for what is written in it but for who signed it. Even I myself may, perhaps, stumble upon the right book and believe it is me on accumulating evidence. What is impossible is the first-person closure: to know, with full grounding, "this is my law and it is valid." The prohibition is not on the content; it is on my relation to it. And this is an astonishing

result for formal logic of all disciplines - the discipline built to abolish point of view - to prove about itself that understanding carries an ineradicable structure of point of view: every system sees what is beneath it completely, and no one sees himself.

The second refinement: the barrier moves. The flexibility is never used up - but it also never catches its own tail. I learned from Gödel that my book cannot establish its own reliability? I will add a new rule to it - "the old book is reliable" - and get a new, stronger book. And indeed, the new book proves what the old one missed. But the new book has a new blind spot of its own: a reliability of its own that it cannot establish, a truth of its own that it misses - and I, if I hold it and trust it, will override it too. Turing studied this ladder - book upon book upon book, without end - and showed that it never closes (Appendix C). Every floor has a ceiling; climbing does not abolish the ceiling, only replaces it with a higher one. At every stage the flexibility is whole and unspent; at no stage does it overtake itself. The blind spot is not a place; it is a relation you stand in to yourself, and it is preserved under every self-improvement you are capable of performing. Like the shadow cast by the lantern you are holding.

F. The Extension: Humanity as a System

So far we have spoken of an individual. But one might object: the individual is limited, while knowledge is a collective enterprise. Mathematics is not what one head contains but what the community, across its generations, establishes - an accumulating tradition, mutual criticism, an archive. Perhaps what the individual cannot do, humanity can?

And indeed, the collective flexibility is even more impressive than the personal. The whole history of science is a chain of overrides: a generation holds a book, trusts it, discovers its limit - and writes a bigger one. Newton was swallowed into Einstein; Euclid became a special case; the foundations of mathematics were reformulated three times in a single century. It seems there is no book humanity cannot rise above.

But exactly here the argument strengthens instead of weakening. Consider the totality of claims that humanity, across its entire future, will ever justifiably establish - every theorem that will be proved, every result that will be verified, every override that will be overridden, from today to the end of all generations. Everything that enters this pile is transmitted in finite strings of symbols and accumulates in an archive. If every phenomenon falls under law, the process that produces the pile falls under law too - and then the pile as a whole, with all its overrides, is exactly the kind of thing Gödel's theorems apply to: one great rulebook, written not on a shelf but on the move, across generations. And still a book. (The precise version of this step is in Appendix D.)

And from here everything repeats, one level up. There exists a truth that humanity as a whole, in all its generations, will never establish - and first among them: that our

collective thinking does not contradict itself. And the trap returns: if humanity is a book, it cannot both hold the book and trust it. **A theory-of-everything of the human mind - a perfect cognitive science that humanity writes, verifies, and declares "this is our law and it is correct" - is an object whose existence contradicts itself.** Had we held it, we would have overridden it by the very holding - refuting the declaration we had just made.

The objector will say: but that is exactly what you showed - humanity overrides every book! It is not a frozen book. The answer: the overrides themselves are part of the book. Which generation will rise above which book, which rules it will adopt, which concepts it will invent - all these too are produced by the same brains, the same physics, the same law. Our flexibility is included in our description; to override book after book is not to leave the system but to run it. What would have been needed in order to truly leave is growth that follows no law at all. Whoever believes there is such growth in us - is permitted to. But notice that he is no longer disputing the argument: the whole argument opens with "if everything is lawful," and he simply believes that not everything is lawful - that is the first option in Gödel's either-or, and it is as open as it ever was. Only one thing he does not have: evidence from the growth itself. A brain that invents calculus according to law and a brain that invents it according to no law produce exactly the same calculus - the growth looks the same from both sides, and so it testifies to neither. (Appendix D.)

And here the decisive difference between the individual and humanity is revealed, the one that makes the collective version sharper than the personal. The individual, we said, has a way out: the other. What I cannot certify about myself, you can certify about me; all of science is built on this exchange of perspectives, on each of us serving as the upper floor for his fellow. But humanity **has no other**. The whole building - all the books, all the overrides, all the certificates ever signed one floor up - is humanity itself, and above the last floor there is no further floor that anyone stands on. There is no external community to serve as its third person; no rung from which its contour could be seen. The individual's blind spot is visible from somewhere else; the blind spot of the human species is a blind spot **that has no "somewhere else."** If there exist truths our law does not reach - our collective consistency first among them - they are hidden not as the next mountain is hidden from the hiker, but as what lies outside the light cone is hidden from the physicist: structurally, from every position we can ever occupy.

So the full statement of the thesis is this: the flexibility to override any book, joined to the intuition that everything falls under law, entails that humanity will never hold its own lawfulness - not because of technological limitation, not because of the brain's complexity, but because of a logical theorem: any formulation we could present and certify, we would override by the very act of certifying, and therefore it is not the formulation.

G. The Top of the Ladder: The Book That Needs No Floor

One question remains, and it is the strangest of all: what could possibly close the ladder? What kind of book could contain the sentence "I am reliable" - and be reliable?

What we have seen so far narrows the options brutally. An ordinary book - mechanically checkable, arithmetic-knowing - that declares its own reliability is not reliable; the declaration itself is the proof. A book that gives up mechanical checkability in order to escape the verdict ceases to be a book: it cannot be handed over, cannot be taught, nothing can be checked against it. One option seems to remain, the most magnificent of all: the complete book - the one containing all truths whatsoever, a consistent book that covers reality to its very end. And here is the surprise Tarski hid at the top of the ladder: even this book does not contain its own certificate of integrity. Not because it is false - because it cannot be phrased in its language. "Everything written here is true" is said always one floor up; even the book-of-everything has a floor it does not reach - the floor from which one speaks about it. Coverage is not enough. A book that covers everything would have had to cover its own covering too, and the covering of the covering, and so on - and this regress closes in no language. (Details in Appendix A.)

The conclusion is precise and strange: the only entity that could declare "I am reliable" and stand by its word is not a bigger book. It is a different kind of thing altogether - one for whom saying and doing are one. And honesty requires stating how it stands by its word: not because its word was checked and found faithful, but because its word cannot fail to come true. The demand for a certificate is not met there; it evaporates there. A description, however broad, always needs someone to sign for it from outside; the only one who needs no signature is the one who does not describe reality but constitutes it. A book in which is written "Let there be light" - and the light does not wait for approval. At the top of the ladder no book stands any longer; what stands there, if anything stands there, is what the tradition calls the utterance by which the world was created.

And one more honesty, more important still: even if such a top exists, its closure does not come down the stairs. Speech that constitutes its own truth is closed and grounded - for the one who speaks it. Whoever stands one floor below and hears it receives a declaration whose truth is guaranteed by the nature of the speaker; and in order to be convinced of that guarantee, he must already trust the nature of the speaker. Nothing of the closure passes downward as proof. It passes only as trust. The asymmetry that has accompanied us all the way up the ladder survives at infinity too: the only one who can vouch for God is God. And perhaps that is why the word the tradition kept for the relation to the top of the ladder is not "knowledge" but faith - not as a surrender of the intellect, but as the only form in which the certificate of the top floor can be received at all.

And this casts a backward light on the intuition we started from. "Everything can be explained by law" means: there exists, in principle, a book that covers everything. But a book that covers everything and cannot sign for itself is not what the intuition asked for - it asked for a law that is complete, certified, reliable, one you could stand on. And what answers that demand, we have proved, is no book of any kind. The intuition of total lawfulness never pointed at a book; it pointed, without knowing it, at something beyond every book. The belief that the world is lawful to its very end was, from the beginning, a theological belief in disguise.

(Intellectual honesty also requires marking the boundaries of the argument - four caveats: where it holds, and what it does not prove. They are collected in Appendix F.)

H. Conclusion: The Law That Holds Precisely by Hiding

Some want to read the Gödel family of arguments as a refutation of the scientific picture of the world - a proof that mind exceeds matter. We have seen that this reading does not stand: the theorems do not decide the disjunction, and Gödel himself refused to decide it. Our flexibility is real, but it overrides every given book - not the law itself. But the opposite, dismissive reading - "nothing here but a logical curiosity" - is just as wrong. There is a real theorem here about our place: **determinism about the mind is consistent with logic; transparent determinism is not.** The law is permitted to hold; what is not permitted to exist is the law held in hand - formulated, certified, and exhaustive, all at once.

And the deep irony is that the flexibility does not weaken the intuition of lawfulness but **completes** it. A law whose power suffices to cover the law-formulator too is exactly a law whose power suffices to hide from him: it must produce creatures who override every book presented to them - and thereby ensure that its own book is never presented. Total lawfulness, the kind nothing escapes, must count among its consequences the theorem of its own invisibility - otherwise it would leave outside itself at least one fact: the fact of its own exposure. The humanity that formulates the laws of everything will discover, at the end of all the formulations, that the final sentence - "and this is our law, and it is complete and correct" - is the only one whose ink never dries. The eye that proved it cannot see itself - and proved this theorem too - is the most faithful witness we have that we are, perhaps, made of law after all.

And if we are made of law - it is of the kind at the top of the ladder: not a book lying on a shelf waiting to be opened, but an utterance we cannot quote because we are the quotation. The declaration "I am reliable" is invalid in the mouth of every book, and valid in one mouth only - the mouth whose saying does the thing. Everyone else gets their signature from the neighbor one floor up.

Appendix A. The Formal Background: the Incompleteness Theorems, Löb's Theorem, and Tarski's Theorem

This appendix states precisely the results that Sections C and G present in words. The "book" of the main text is the formal system F below.

In 1931 Kurt Gödel proved his two incompleteness theorems (Gödel 1931). We need precision here, because the entire force of the argument depends on it.

Let F be a formal system satisfying three conditions: it is **consistent** (proves no contradiction); it is **recursively axiomatizable** (there is an algorithm that recognizes what is an axiom and what is a valid inference - that is, its set of theorems is computably enumerable); and it **contains basic arithmetic** (a tiny fragment suffices, Robinson's system Q). Then:

The first theorem: There exists a sentence G in the language of F such that F proves neither G nor its negation. (A precision: from consistency alone it follows only that G is unprovable; to rule out the provability of the negation as well, Gödel's original formulation needed a slightly stronger condition - omega-consistency - and Rosser showed how to make do with consistency alone, at the price of replacing the sentence with a cleverer one (Rosser 1936). In what follows we need Gödel's original sentence specifically, whose power lies in its equivalence to consistency.) Moreover - and this is the point most popular discussions miss - F itself can prove the conditional claim: "if I am consistent, then G is not provable in me." That is, the system knows about its own blind spot; what it cannot do is discharge the condition.

The second theorem: F does not prove its own consistency, $\text{Con}(F)$. In fact F proves the equivalence between $\text{Con}(F)$ and G - so a self-proof of consistency would entail G, contradicting the first theorem. Worse: F proves that "if I prove my own consistency, I am not consistent." A certificate of integrity that the system issues to itself is evidence against it.

Why does the self-declaration shatter the system? The chain has three steps. (a) The first theorem, when formalized inside F itself, yields: F proves the claim "if $\text{Con}(F)$, then G is not provable in me." (b) But "G is not provable in me" is, word for word, the content of G; so F proves: if $\text{Con}(F)$ then G. (c) Now suppose F proves $\text{Con}(F)$. Then F proves G. But F has mechanical access to its own proofs: from the existence of a proof of G it follows that F also proves "G is provable in me" - which is exactly the negation of G. So F proves G and its negation: contradiction. The declaration "I am consistent" is not a description of the system but one more theorem inside it - and it is exactly the missing link for completing a contradiction of which the system already carries half.

It is important to emphasize the role of the three conditions, because each is sharp: Presburger arithmetic (addition only) is complete and decidable; "true arithmetic" (the

set of all true sentences about the natural numbers) is complete by definition - but not recursively axiomatizable, and therefore not a "system" you could give to anyone; and an inconsistent system proves everything. Gödel's theorem is thus a trilemma: **mechanical checkability, enough arithmetic, completeness - pick two.**

And one more precision about the third condition: the tiny fragment Q suffices for the first theorem only. The reflective claims - F's ability to prove the first theorem about itself, the second theorem, the chain above, and Löb's theorem below - require that F also verify three basic facts about its own provability predicate. These three facts are called the Hilbert-Bernays-Löb derivability conditions (Hilbert & Bernays 1939; Löb 1955):

The first condition - the system recognizes its own proofs. If F actually proves P, then F also proves the sentence "P is provable in me." The reason: a proof is a finite object, checking it is a computation, and a system that knows arithmetic can follow that computation and certify it. (In step (c) of the chain above we used exactly this: once F proved G, it also proves "G is provable in me.")

The second condition - the system knows that proofs concatenate. F proves: "if 'P implies Q' is provable in me and P is provable in me, then Q is provable in me." Whoever holds a proof of an implication and a proof of its antecedent holds a proof of the consequent - the gluing of the two proofs - and the system can certify the fact of the gluing as well.

The third condition - the first condition, known to the system itself. F proves: "if P is provable in me, then 'P is provable in me' is provable in me." It is not enough that the system recognizes its proofs in practice; it also proves about itself that it is such. This is the subtlest of the three - the difference between doing a thing and knowing that you do it.

On exactly that difference Q falls: it recognizes its proofs in practice (the first condition holds of it as an external fact), but it is too weak to prove the second and third conditions about itself. Every reasonable system from elementary arithmetic upward verifies all three, and certainly Peano arithmetic.

And why exactly these three and not some other list? Because it turns out they exhaust the story: Solovay's theorem (Solovay 1976) establishes that for systems like Peano arithmetic, everything the system ever proves about its own "provable in me" - at the level of sentence combinations - follows from the three conditions together with Löb's theorem, and from nothing more. The book's entire self-knowledge amounts to three lines and one ceiling.

One more tool is needed: Löb's theorem. Its origin is a question posed by Leon Henkin in 1952 (Henkin 1952). Gödel's sentence says "I am not provable in me," and it turned

out to be true and unprovable - it keeps its word. Henkin asked: what about its inverted twin, the sentence that says "I am provable"? Is it provable, in which case it spoke the truth, or unprovable, in which case it is a stable lie that the system never refutes? Intuitions split. Martin Hugo Löb answered (Löb 1955): the sentence is provable. And not only that - the answer follows from a far more general rule: **if F proves the claim "if P is provable in me then P," then F already proves P itself.** For any P whatsoever.

In other words: no system can establish even conditional trust in itself - "what I prove is true" - with respect to any sentence not yet settled. Conditional trust sounds cheap: it does not claim that P is true, only that "if I prove it, it will be true." And the theorem says the cheap thing does not exist: the moment the system holds it, the "if" fulfills itself, and the hedge collapses into an assertion. Self-trust is not hard to obtain; it is impossible in a provable way, and the proof is carried out by the very faculty whose reliability is in question.

Why? The proof rests on a helper sentence, L, and it is worth pausing to say what it is - and along the way it will also become clear how G itself was built. How can a sentence speak about its own provability at all? This is the technical invention at the heart of this entire chapter, and it is built from three components. (1) Numbering: every sentence in the language of F receives a serial number - a "Gödel number" - just as every book in a library receives a catalog number. (2) Arithmetization: the property "the sentence whose number is n is provable in F" turns out to be an ordinary arithmetic property of the number n - a claim about numbers, exactly the kind F knows how to talk about. (3) The diagonal lemma: for every arithmetic property that can be phrased, there exists a sentence asserting that property of its own number - and the equivalence between the sentence and the claim-about-its-number is provable in F. The sentence does not say "I"; it names a number, which happens to be its own. There is no mystery here - only arithmetic. The sentence G we met earlier is the product of this construction for the property "is not provable"; L is the product of exactly the same construction for the property "if provable - then P." That is: L is an ordinary arithmetic sentence whose content is equivalent (provably in F) to the claim "if L is provable, then P." And in everyday words: L is a conditional promissory note whose trigger is its own validation - "if you ever prove me, you owe P."

And from here the chain, in three steps. (a) A bookkeeping step, which as yet needs no trust at all: suppose, inside F, that "L is provable." If L is provable, then its content - "if L is provable, then P" - is provable; and the supposition itself, "L is provable," can be certified (the third condition). So we hold a provable implication and a provable antecedent, and by the second condition the consequent is provable: "P is provable." Summary of the step: if L is provable, then P is *provable*. Note that we did not get P itself - from "L is provable" it follows that L's content is provable, not that its content is

true; bookkeeping about proofs always stays one floor down. (b) Here, and only here, the trust enters: the assumption "if P is provable then P" upgrades the conclusion from "P is provable" to P itself. We have obtained: if L is provable, then P. (c) And now look at what we have just proved - "if L is provable, then P." That is, word for word, L itself: we have proved L without noticing. And having actually proved L, F certifies "L is provable" (the first condition) - which is exactly the trigger of the note now in our hands: P. Without the trust, the argument would have stalled at "if L is provable, then P is provable" - true, harmless, and leading nowhere; the trust supplied the final rung, and the note proved itself and then cashed itself.

Two consequences of this rule. The first: trust and proof coincide exactly, in both directions - but the directions are not equal. In one direction stands Löb's theorem: if F proves "if P is provable then P," it proves P - that is the surprise. The opposite direction is cheap: whoever has proved P holds "if P is provable then P" automatically, since he holds something stronger - P itself, with no condition at all. Combining the two directions: self-trust buys the system nothing it did not already own. And the second consequence: the second incompleteness theorem is the special case $P = \text{contradiction}$. $\text{Con}(F)$ is equivalent to the sentence "if a contradiction is provable in me, then a contradiction" - self-trust with respect to the one sentence that must never be provable - and Löb says such trust is available only to a system that already proves a contradiction.

Tarski's theorem (Tarski 1936): The set of (Gödel numbers of) true sentences in the language of arithmetic is not definable in the language of arithmetic; in general, no sufficiently rich language capable of self-reference (in classical logic) contains a truth predicate for itself. Hence the distinction in Section C: consistency - "I prove no contradiction" - the system can phrase (it is an arithmetic sentence, $\text{Con}(F)$), and is only forbidden to prove; full soundness - "everything I prove is true" - it cannot even phrase as a single sentence in its language, but at most as a schema of sentences, and that is blocked by Löb's theorem. Hence also the claim of Section G about the "book of everything": true arithmetic is complete and consistent, but by Tarski it is not definable in its own language - so its certificate of reliability does not exist as an arithmetic sentence; it is written only in a richer language, just as set theory proves the soundness of Peano arithmetic and does not prove its own. Every floor signs for the floor beneath it; no floor signs for itself.

A note on a non-classical way out: Saul Kripke showed that a language can contain a truth predicate for itself, if certain sentences are permitted to remain without a truth value (Kripke 1975). But in Kripke's construction, a global declaration of the form "everything I say is true" remains, in the minimal construction, ungrounded forever: the sentence exists in the language, and never receives a truth value. The signature line is printed - and the ink, quite literally, never dries. The non-classical way out does not escape the phenomenon; it restates it.

As for Goodstein's theorem of Section C: write a natural number in hereditary base- n notation (the exponents too are written in base n , and the exponents of the exponents, and so on); replace the base n by $n+1$ and subtract 1; repeat the process with an ever-growing base. Goodstein's theorem (Goodstein 1944) states that every such sequence terminates at zero. The standard proof attaches to each term of the sequence an ordinal below epsilon-zero and shows that the sequence of ordinals strictly decreases - and therefore, by the well-foundedness of the ordinals, the sequence reaches zero. Kirby and Paris (Kirby & Paris 1982) proved that the theorem is not provable in Peano arithmetic: the proof requires transfinite induction up to epsilon-zero - a strength lying exactly beyond its boundary. A result in a similar spirit in finite combinatorics: the Paris-Harrington theorem (Paris & Harrington 1977).

The dictionary matching the language of the main text: "the book does not contradict itself" - consistency; "mechanically checkable" - recursive axiomatizability; "knows arithmetic" - containing Q; "the blind spot" - the sentence G; "the certificate of integrity" - $\text{Con}(F)$; "partial, conditional trust" - Löb's condition; "the signature line that is not printed" - Tarski's theorem; "the override" of Sections B-D - the passage from F plus its soundness to $\text{Con}(F)$; "ordinary arithmetic truths" in the blind spot - Goodstein's theorem and its family; "a floor above" - a system that proves the soundness of its neighbor below, as ZFC proves the soundness of PA.

Appendix B. Deriving the Trilemma

This appendix establishes the central claim of Section D: that the three assumptions cannot hold together. "There is a law" is (1) below, "it can be found" is (2), "it is reliable" is (3).

Assume three things:

(1) The lawfulness assumption: My capacity for mathematical justification is identical to some formal system F (consistent, recursive, containing arithmetic - in the precise sense of Appendix A).

(2) The formulation assumption: I am able to formulate F - to write down its axioms - **and to know** that this formulation is indeed me.

(3) The validity assumption: I know that F is sound - that its theorems are true.

From the three assumptions a contradiction follows, in three steps: from (2) and (3) I may assert, with full justification, $\text{Con}(F)$ - since I know which system I am, and know that it proves no falsehood, and a sound system is in particular consistent. But by the second incompleteness theorem, F does not prove $\text{Con}(F)$. So I have established a claim beyond F's reach - contradicting (1), which says that F exhausts my capacity.

The three assumptions cannot live together. That is the theorem. But the theorem does not say which of them is false.

Appendix C. The Ladder That Never Closes, and the Epistemic Vindication

This appendix establishes two claims from the main text: the "moving barrier" of Section E, and the formal vindication of the thesis mentioned in Section D.

The ladder. Given F , one can adopt $\text{Con}(F)$ as a new axiom and pass to $F' = F + \text{Con}(F)$. This move settles the old Gödel sentence: F' proves the G of F . (The step requires F to be sound, not merely consistent: a consistent system that proves the negation of its own consistency - and such systems exist - collapses precisely upon the addition of its Con . If F is sound, F' is sound too, and the theorems apply to it.) But F' has a Gödel sentence of its own, a consistency of its own - $\text{Con}(F')$ - unprovable in it, and a contour of its own. Turing studied the transfinite progressions of theories obtained by repeating this step along ordinals (Turing 1939), and Feferman developed the theory in full (Feferman 1962): the ladder never closes, and the decision as to which ordinals are "well-founded" - that is, along which stages one may climb at all - is itself a decision the climbing system cannot fully establish. This is the precise content of the sentence "the barrier moves": self-improvement is possible, but it duplicates the blind spot instead of erasing it.

The epistemic vindication. William Reinhardt showed, in the framework of epistemic arithmetic (an arithmetic language enriched with a knowledge operator), that the claim "I am *some* Turing machine" is consistent with everything an ideal knower can know, while the claim "I know *which* Turing machine I am" (together with knowledge of one's own soundness) is refutable (Reinhardt 1986). Timothy Carlson completed the consistency proof (Carlson 2000). This is, in formal dress, exactly the distinction Section D insists on: the first assumption alone - consistent; the conjunction of all three - refuted.

Appendix D. Humanity as a System: the Precise Version

This appendix establishes the generalization step of Section F - from the individual to humanity.

Let H be the set of all mathematical claims that humanity, across its entire future, will justifiably establish. The decisive question: is H computably enumerable - that is, is there an algorithm that enumerates its members - thereby satisfying the conditions of Gödel's theorems?

The lawfulness assumption entails that it is: the number of human beings at any moment is finite; every proof ever written is a finite object; the chain of generations evolves according to the laws of physics, and the output of a computational process is a computably enumerable set. H is therefore the set of theorems of a single formal system - F-humanity - unrolling in time. And on the assumption that the pile is consistent - that humanity never justifiably establishes a contradiction - the second incompleteness theorem applies to it: $\text{Con}(\text{F-humanity})$ is true and is not among its theorems. (Without the consistency assumption there is no conclusion - a contradictory system "proves" its consistency too, worthlessly; but whoever assumes that humanity justifiably establishes contradictions has already given up far more than this argument.) This is "the truth that humanity will never establish" of the main text.

As for the growth objection: the sequence of reflections humanity will perform - which consistency principles it will adopt, which ordinals it will recognize as well-founded, when and how - is itself produced by the same lawful process, and so the closure of all our possible types is included in F-humanity from the start. Exceeding F-humanity requires a non-computable step, and here the objection splits in two. If the objector remains within the lawfulness assumption, his growth is computable - included in F-humanity from the start - and the objection has been answered. And if he proposes growth that follows no law at all, he is no longer objecting to the argument but abandoning its premise: the thesis of the essay is conditional - *if* everything is lawful, the law will not be held - and whoever denies the antecedent does not refute the conditional; he chooses the first horn of Gödel's disjunction, "the mind surpasses every machine."

Appendix E. The Probabilistic Caveat

This appendix establishes the second caveat of Appendix F.

The diagonal argument - in its logical version as in the predictor version of Section B - rests on inverting a definite output: one takes what the system says and does the opposite. For that, the output must be unambiguous. A stochastic system supplies no such output - you cannot do the opposite of a coin flip. And indeed, the literature contains "reflective oracles" that demonstrate consistent self-prediction at a probabilistic price: they repair the diagonal point by randomization, and a fixed point (instead of a contradiction) is guaranteed by compactness considerations (Fallenstein, Taylor & Christiano 2015). Hence the cautious phrasing in the main text: if our lawfulness is probabilistic at bottom, only certain self-knowledge is excluded.

Appendix F. Caveats, So That No More Is Claimed Than Was Proved

This appendix marks the boundaries of the entire argument, which are four: where it holds, and what it does not prove.

The first caveat - the idealization. "What humanity is capable of establishing" is not a sharp concept: it abstracts from memory, from lifespan, from errors, from the future death of the species (Feferman 2006; Horsten & Welch 2016). The argument holds under this abstraction only. But the caveat cuts in both directions: the claim that the mind surpasses every machine, and the lawfulness assumption itself, are formulated at the same level of abstraction. The trap stands firm under any sharpening you choose.

The second caveat - randomness. The override requires a book that speaks plainly: to stand on a book and best it, you must know what it says. A law that flips coins escapes the mechanism: you cannot "do the opposite" of what a coin will predict. The all-predicting computer of Section B, had it announced a probability distribution instead of a single act, would have been immune to my circumvention. And indeed, there exist theoretical systems that demonstrate consistent self-knowledge at a probabilistic price (Appendix E). If our lawfulness is probabilistic at bottom, the barrier refines: not all self-knowledge is excluded, only certain self-knowledge; probabilistic self-knowledge remains open.

The third caveat - the meaning of the word "formulate." The argument applies to laws that can be written and checked: for which there is a mechanical procedure deciding what follows from them and what does not. A law that is not like this escapes the mechanism - but it is also not a "formulation" in any useful sense: it cannot be handed over, cannot be taught, no proof can be checked against it. Whoever proposes that our lawfulness exists but cannot be written this way has already given up the intuition he set out to save, for "explanation by law" has always meant an explanation that can be handed over.

The fourth caveat - what the top of the ladder does not prove. Section G is not an existence proof. It is a classification claim: if there exists something that closes the ladder, it is not a book. Nothing in this is evidence that the ladder actually closes - it may simply go up and up, with nothing at its top. What was proved is only this: whoever believes in a lawfulness that is complete and certified believes, willingly or not, in something that is not a book.

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